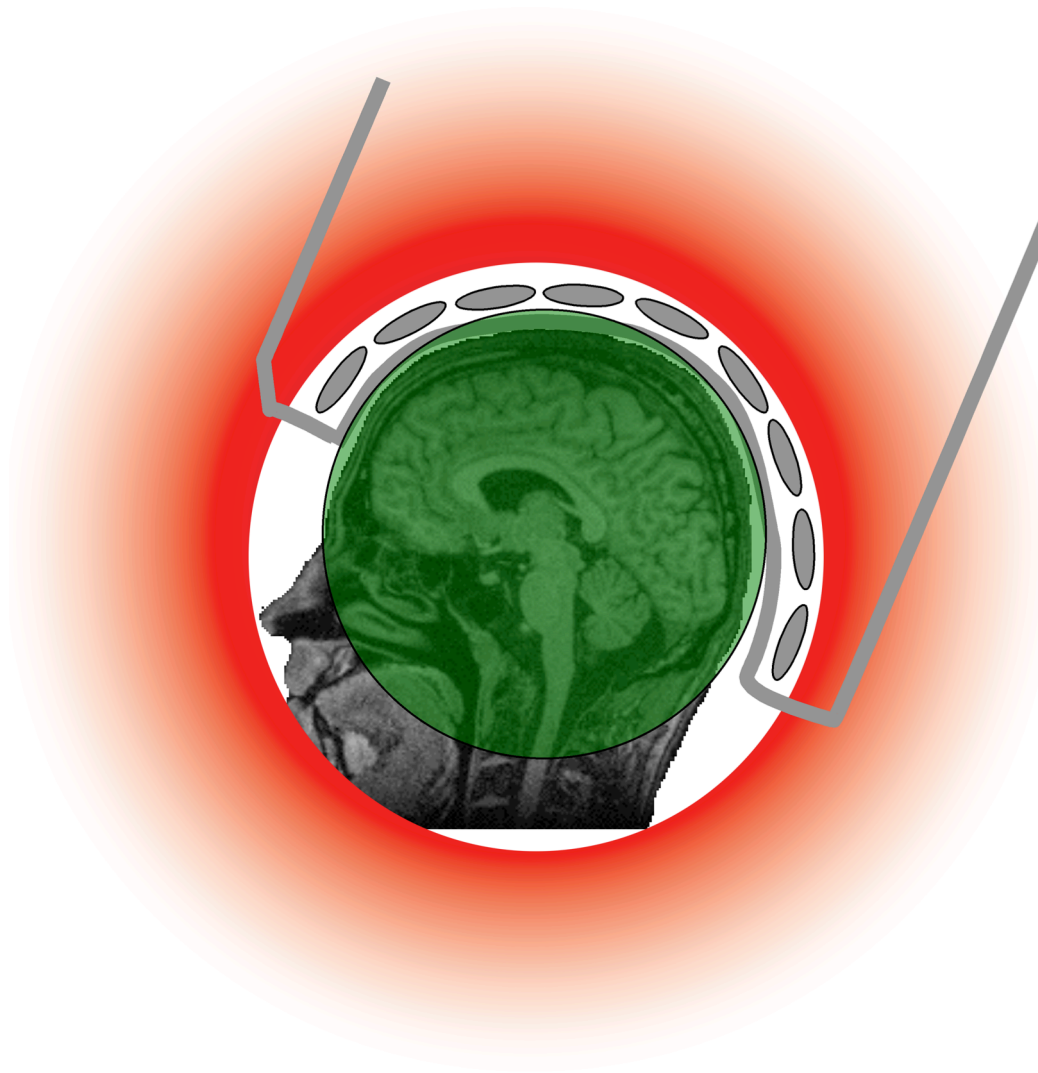


Elekta Neuromag®

Addendum to MaxFilter™ 2.2 User's Guide: Individual MEG Channel Artifacts — A Case Example



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Manufacturer:

Elekta Oy

Siltasaarencatu 18-20 A

FI-00530 Helsinki, Finland

Tel: +358 9 756 2400

Fax: +358 9 756 24011

Web: www.elekta.com

E-mail: meg-support@elekta.com

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Cautions and notes



Warning: Warnings are directions which, if not followed, could constitute a health hazard, cause fatal or serious injury, or lead to erroneous clinical diagnosis and possible to clinical mistreatment.



Caution: Cautions are directions which must be followed in order to ensure safe and efficient operation and to avoid damage to system.

Note:

Notes provide advice and recommendations for safe and efficient use of the product as well as highlight unusual points.

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1 Overview

1.1 Scope

We are providing the information in this Addendum to notify you of an important safety issue that may exist on your equipment, and to inform you of any actions needed to safeguard both your staff and your patients. We ask that you please read and understand the content of this Addendum and implement any recommendations provided.

We advise you to insert this Addendum in the applicable copy of the User Manual.

1.2 Description

Although there was no injury or harm to the patient in the described case and there are already instructions for safe use in place for the MEG system users, we want to take precautionary measures and inform our customers to be aware of the following:

This Addendum presents a user-reported case example in which the automatic methods for detection and removal of malfunctioning channels implemented in MaxFilter™ failed to recognize an abrupt artifact on a single MEG channel. In this case example, MaxFilter™ spread the channel artifact on multiple channels in the neighborhood of the affected one, an expected result of most spatial filtering techniques, including signal space separation (SSS).

The user was able to identify the problem by carefully inspecting the spatially-filtered data against the original data, and corrective measures were subsequently taken. There were no adverse consequences affecting the patient care.

However, this case represents the worst-case scenario, in which the artifact is particularly difficult to recognize. In typical cases, the output data produced by MaxFilter™ in response to single-channel artifacts has waveforms immediately recognizable as artifacts, and the source is localized in a clearly non-physiological location. In other words, these artifacts are typically easily detectable by a trained physician. In this particular case example, however, the output waveforms resembled epileptiform discharges after temporal filtering, making them particularly difficult to detect.

This Addendum clarifies the guidance provided by the Instructions for Use, providing specific guidance on how to inspect the output data in order to detect similar artifacts.

1.3 Clinical impact

Though the artifact can be recognized by existing procedures, there is a very small chance that the spreading of the channel artifact would lead to erroneous conclusions about the epileptic brain activity.

1.4 Solution

It is emphasized that it is important to inspect the raw data to which MaxFilter™ processing was applied. This is described by the user manual of MaxFilter™ as well as by the Clinical Practice Guidelines issued by the American Clinical MEG Society (ACMEGS).

Should an activity only be present on a single MEG channel in the raw data, it is most likely a channel artifact and must thus be disregarded.

Moreover, even if the MEG analysis is accepted by the user, the result must not be used alone for the pre-surgical planning but must only be used in combination

with other supportive diagnostic data, as described in our Intended Use and Indications for Use statements.

1.5 Technical Reference

NM24057A-* MaxFilter™ User's Guide (non-US version)

NM24124A-* MaxFilter™ User's Guide (US version)

2 Case description

2.1 Safety reference

American Clinical Neurophysiology Society (ACNS) and American Clinical MEG Society (ACMEGS) guidelines consider the visual inspection of the raw data as an obligatory step in MEG-EEG recording and analysis of spontaneous activity [1]:

“Waveforms of MEG and EEG (“raw data”, original data as collected) for the entire recording should be visually examined, following the principles established for clinical EEG. Visual inspection of time series is an obligatory initial step in the analysis of spontaneous MEG–EEG data that is aimed at the (1) identification of artifacts, (2) evaluation of overall data quality and integrity, and (3) identification of background rhythms, asymmetries and other background characteristics, and IIEDs, including morphologic and temporal characteristics, in both MEG and EEG. These findings should be evaluated and reported systematically for each study.”

The following warnings and cautions are associated with this Addendum (see MaxFilter™ User’s Guide):



Warning: It is important that the user inspects both the input and the output data visually to judge the quality for the *MaxFilter* result.

2.2 Introduction

This case example shows an adolescent patient with a history of intractable seizures, resective brain surgery, and an implanted VNS.

The user identified several suspected interictal epileptic discharge (IIEDs) waveforms in the post-processed data (tSSS processing using MaxFilter™). Figure 1 shows the results of the analysis of one such waveform. The activity is shown on multiple MEG channels (see the time traces at lower left panel of the figure), and the transient duration is shorter (<40 ms) than real epileptiform spikes. No activity is seen in EEG (see the time traces at the bottom left corner of the figure). The artifact activity seems to originate from the right mesial frontal area.

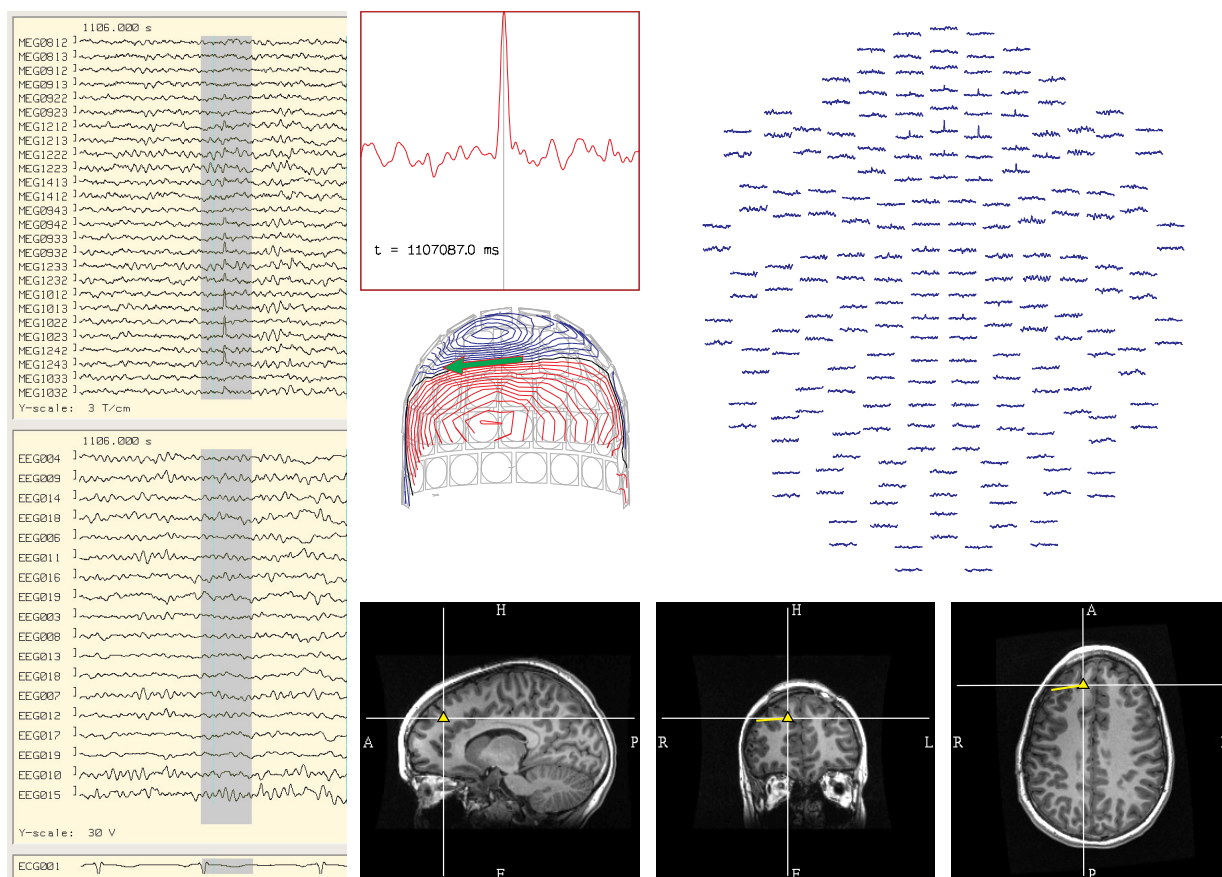


Figure 1 The analysis performed on frontally localized activity in interictal MEG recording. The time traces in the left upper panel show the frontal MEG channels where the activity is most pronounced. The time traces in the left lower panel show all the EEG channels in longitudinal bipolar montage. The EEG montage does not show any activity.

2.3 Inspection of raw data

We describe here the steps needed to inspect whether the activity seen in Figure 1 could be artifact or not, using the Graph module of the Elekta Neuromag® Data Analysis software.

The steps assume the usage of the “basic.setup” Graph configuration. A similar procedure can be followed, when using other Graph setups.

1. Open **Graph** through the **Neuromag** menu in the operating system start menu.
2. In the **Graph** main menu, select **File->Load settings**.
3. Select folder **examples**, and press **Filter** button or double-click the folder name.
4. Select file **basic.setup**, and press **OK** button or double-click the file name.
5. In the Graph menu, select **File->Open Diskfile**.
6. Navigate to the folder, where your MaxFilter™ processed file is located in.
7. Select the file, and press **OK** button or double-click the file name.

The file is loaded in, and a subset of channels is displayed on the screen.

8. Localize the time region and the channels where your activity of interest is located.

*You can navigate the MEG data along the time axis by clicking the backward or forward arrow button of the slider at the bottom of the screen. This will move the time window backward or forward one step. You can click in the dark gray empty space between the slider rectangle and the arrow buttons to move forward or backward one page at a time. You can also go to a specific time point by typing time (in seconds) to the upper box inside the **Start & length** group in the right-hand side of the Graph main window and pressing enter.*

*You can navigate the MEG channels by opening the **Displays->Selection** menu in the Graph main menu. You can tear it off the main menu by pressing the hatched line at the top of the **Displays->Selection** (just above the **Device** menu item). Selecting any of the channel groups in the **Displays->Selection** menu will update the display to show the subset of channels defined by the selection. The channel selections are divided to those for gradiometer channels (top part of the menu), and those for the magnetometer channels (bottom part of the menu, ending with 'mags').*

9. You can zoom in to see the activity better, using two different methods:
 - a. Press right mouse button down at the desired end or start time point, and dragging the mouse until you have selected the time interval you want to see zoomed in the whole window. Select **Displays->Zoom in** from the Graph main menu to zoom in to this portion of the data.
 - b. Change the length of the time interval shown on the screen by typing the desired length of the time window (in seconds) to the lower box inside the **Start & length** group in the right-hand side of the Graph main window and pressing enter.

Figure 2 shows the spike displayed in report of the Figure 1 as inspected in Graph based on the instructions given above.

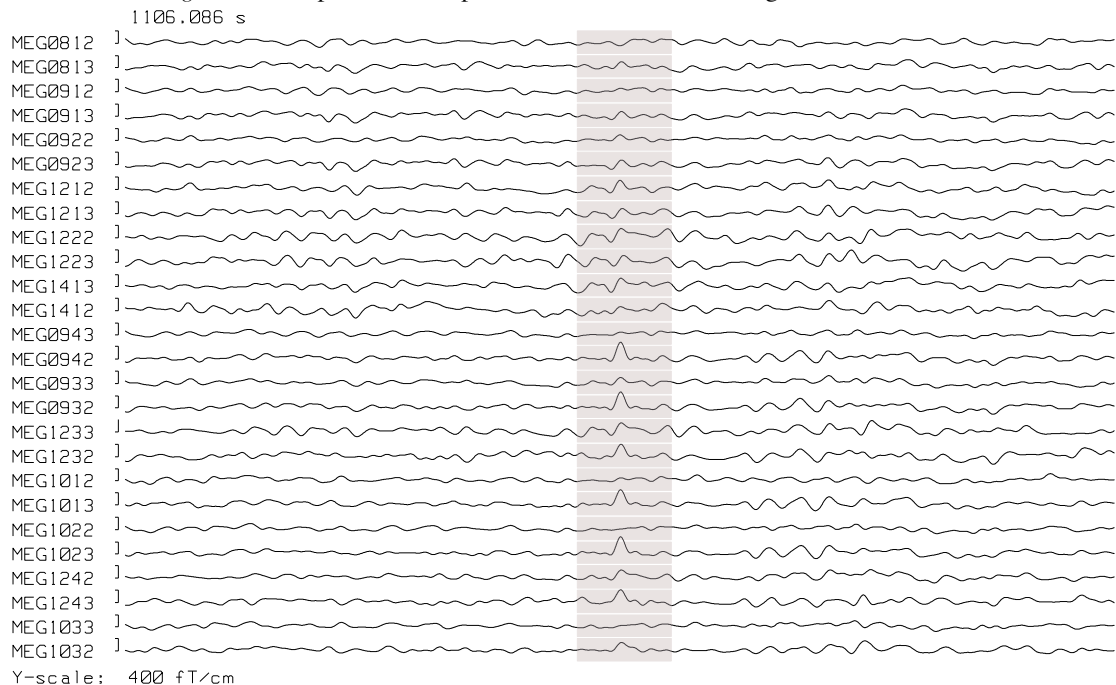


Figure 2 The area of interest in the data from Figure 1 displayed in Graph, according to instruction in steps 1-9. The display shows traces over 2 seconds of data, filtered with 40 Hz temporal low pass filter and DC offset removal.

10. To turn off the temporal filtering, which is by default on in this setup: in Graph main menu, select **Displays->Control Panel**. In the **Control Panel** window, connect **buffer** widget to **meg** widget by-passing the **meg-filter** widget. You can do this by pushing down mouse right button on top of the **buffer** widget (it gets pink), and then dragging the line from **buffer** to the **meg** widget.

Figure 3 shows the MaxFilter™-processed data without any temporal filtering. From the Figure, you can see that temporal evolution of the activity does not resemble an epileptic spike anymore.

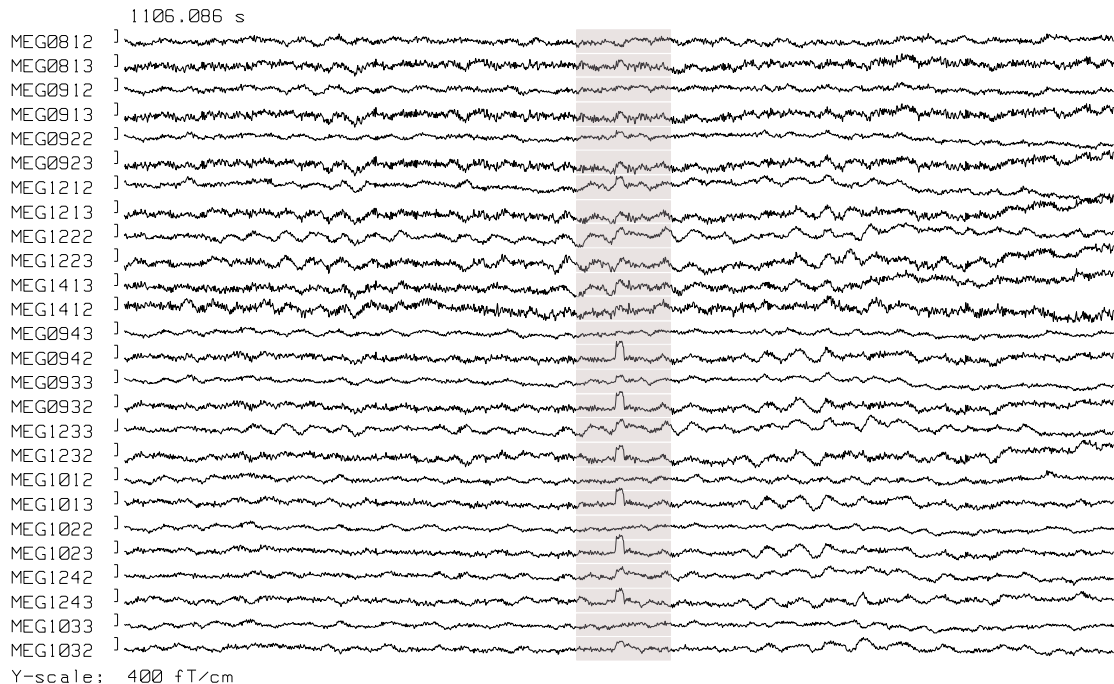


Figure 3 The data shown in Figure 2 displayed after the temporal filtering is turned off according to the instructions provided in step 10.

11. To turn the temporal filtering on again, connect the **meg-filter** widget back to **meg** widget. You can do this by pushing down mouse right button on top of the **meg-filter** widget, and dragging the line forming to the **meg** widget.
12. Close down the **Control Panel**, by selecting **Edit->Close** in the **Control Panel** main menu.
13. To inspect the raw data prior to MaxFilter™ processing, select **File->Open Diskfile**.
14. Select the non-MaxFilter™-processed data file, and press **OK**. The raw data file will be opened from the same time point, and the same channel subset will be displayed that was used for showing the MaxFilter™ processed data.

If the data is from a MaxShield™ system and internal active shielding was turned on when measuring the data, Graph will warn about not opening such a file before MaxFilter™ processing. Continue loading the data anyway for inspection.

Figure 4 shows the data from the case example prior to MaxFilter™ processing. By looking at the raw data, it is evident that the activity arises from an artifact in a single channel. The name/number of the channel is MEG1023. Take note of the channel number.

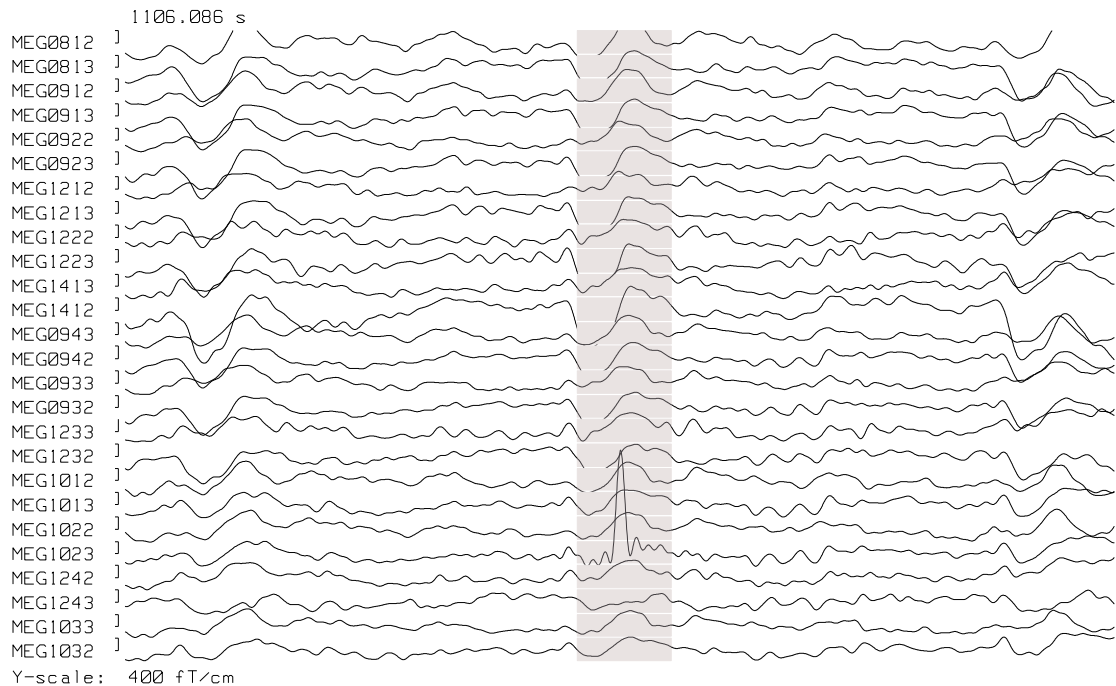


Figure 4 The raw data for the case example prior to MaxFilter™ processing, analyzed as instructed in steps 13-14.

2.4 Manually omitting channels from MaxFilter™ processing

When you have found an artifact in a single channel, you can simple omit this activity from your analysis. However, better way is to re-run MaxFilter™ processing by omitting this channel. The data from this channel will not be used in MaxFilter™ processing, but instead the data will be reconstructed using the signal space separation method.

The steps needed to re-run the MaxFilter™ processing are (we assume here the channel MEG1023 as the channel with artifacts):

15. Open **MaxFilter™** through the **Neuromag** menu in the operating system start menu.
16. Press **Hide** button to hide the copyright and version information notice window.
17. In the **MaxFilter™** main menu, select **File->Load data...**
18. Navigate to the folder, where your raw data file file is located in.
19. Select the raw data file.
20. Select the desired **MaxFilter™** operations from the window, and press **OK** button.
21. In the **MaxFilter™** main menu, select **File->Output options...**
22. Change in the **output file** text box the **.fif** suffix to, e.g., suffix **_badch1.fif**.
23. Press **OK** button.
24. In the **MaxFilter™** main menu, select **Parameters->Bad channels...**
25. Type in the channel number (or numbers) to the text box **List of logical MEG channel numbers** (see Figure 5 for reference).

26. Press **OK** button.

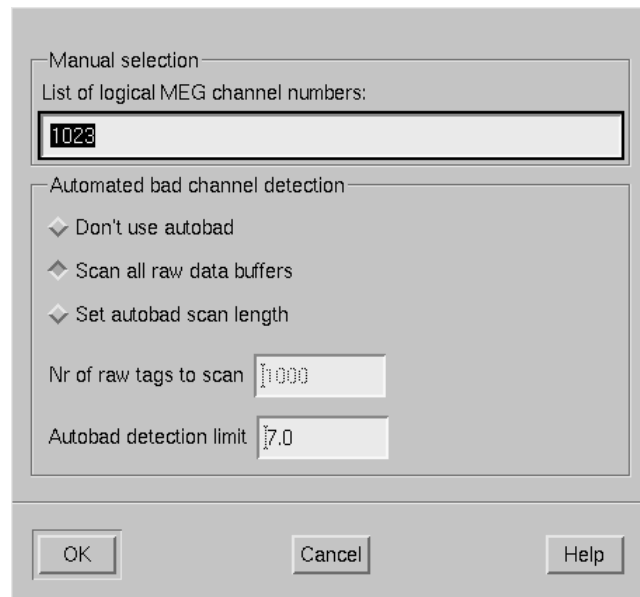


Figure 5 The **MaxFilter™** bad channel selection window. Type in the channel number to the text box **List of logical MEG channel numbers**.

27. In the **MaxFilter™** main menu, select **Parameters->Origin...**
28. Type in the origin settings from your previous analysis, or press the **Fit to isotrak points** button.

Note: it is preferred that the origin is fitted using the MEG-MRI Integration (a.k.a. Mrilab) software. Please refer to the MEG-MRI Integration software for details on how to fit a sphere to a set of points.

29. Press **OK** button.
30. In the **MaxFilter™** main window, press **Execute** button.

MaxFilter™ should start processing. However, if you see a **red label** appear next to the **Execute** button, with text **ERROR!**, something has gone wrong. You can check any warning or error messages by selecting **File->Show log...** in the **MaxFilter™** main menu.

When the **MaxFilter™** processing has ended, the program informs that **Processing finished OK!** under the processing slider at the bottom of the **MaxFilter™** main window, and that it Wrote successfully the result file in the status bar at the bottom of the **MaxFilter™** main window.

31. Inspect the data in Graph, by selecting **File->Open Diskfile** in the Graph main menu.
32. Navigate to locate the processed file (suffix `_badch1.fif`).
33. Press **OK** button.

You will now see the data after the **MaxFilter™** processing with the artifactual channel omitted from the process. Figure 6 shows this the processed data for this case example. The channel artifact has been suppressed from the data.

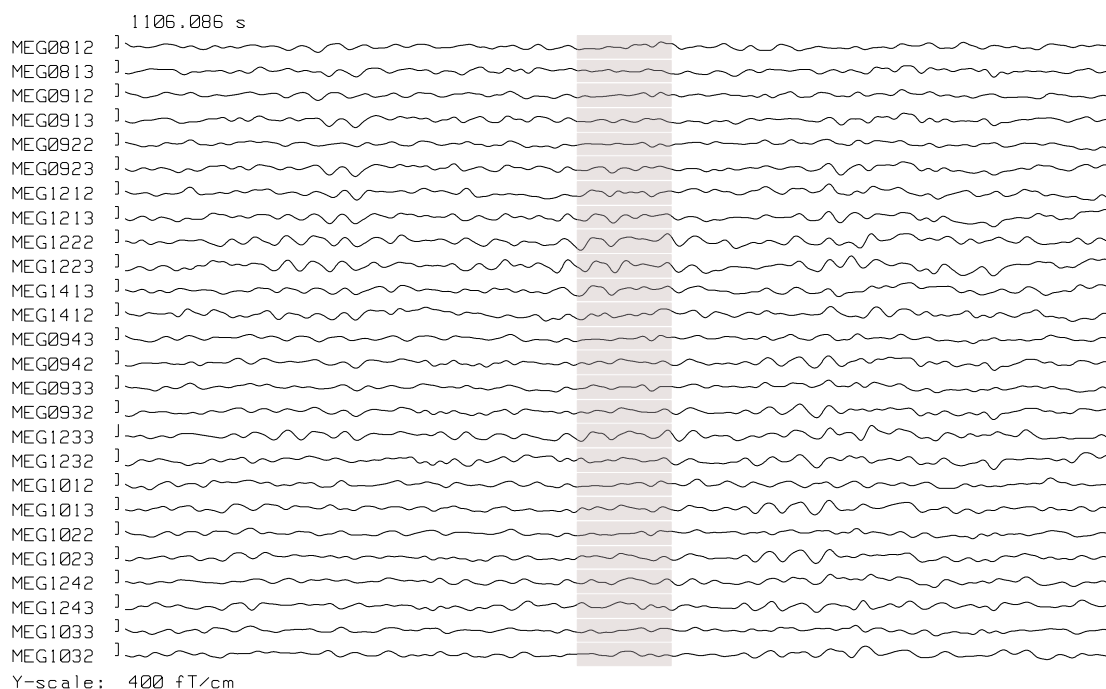


Figure 6 The data of the case example after MaxFilter™ processing was re-run by omitting the channel MEG1023 from the processing. The artifact has been suppressed properly.

2.4

References

- [1] A.I. Bagic, R.C. Knowlton, D.F. Rose, and J.S. Ebersole, for the ACMEGS Clinical Practice Guideline (CPG) Committee, “American Clinical Magnetoencephalography Society Clinical Practice Guideline 1: Recording and Analysis of Spontaneous Cerebral Activity”, J Clin Neurophysiol, 28: 348–354 (20)

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Corporate Head Office:
Elekta AB (publ)
Box 7593, SE-103 93
Stockholm, Sweden

Tel +46 8 587 254 00
Fax +46 8 587 255 00
info@elekta.com

Regional Sales, Marketing and Service:

North America
Atlanta, USA

Tel +1 770 300 9725
Fax +1 770 448 6338
info.america@elekta.com

Europe, Latin America,
Africa, Middle East &
India

Tel +44 1293 544 422
Fax +44 1293 654 321
info.europe@elekta.com

Asia Pacific Hong
Kong, China

Tel +852 2891 2208
Fax +852 2575 7133
info.asia@elekta.com